

DARE: Differential Adversarial Reward Equalization for Humanoid Motion Imitation

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Abstract—Physics-based humanoid imitation must satisfy several heterogeneous objectives at once: global root motion, articulated pose, and velocity all have to agree with a synchronized reference. Handcrafted tracking rewards combine these objectives with weights and scales chosen per motion. Learned differential rewards remove the weights, but they keep the discriminator from collapsing onto its single positive example with a gradient penalty whose strength must again be chosen per motion, because that one coefficient decides both how sharp the discriminator may be and how far typical policy residuals are pushed from the ideal. We introduce DARE, a differential adversarial reward that fixes these two properties separately, with no motion-specific coefficient. The geometry of the critic is fixed by construction: every semantic block of the differential enters through its own spectrally normalized map, so objectives of very different dimension receive equal capacity and equal gain, and the score change relative to the unique zero-error anchor is bounded by the size of the residual. The scale of the critic is fixed once, in closed form, from the reward transform: the balanced adversarial loss places the anchor and the typical policy residual symmetrically around zero, and DARE rescales the critic so that this separation equals the width of the window over which the reward transform responds to the critic’s score. The rule is shared by every motion; only the measured separation depends on the data. We evaluate DARE with one configuration on six humanoid skills with distinct dynamical regimes—Roll, Climb, Backflip, Crawl, Get-Up, and Spin Kick—**[TODO: one-sentence quantitative summary from the formal runs]**, and deploy the resulting controllers on **[TODO: robot platform]**.

I. INTRODUCTION

Physics-based humanoid control must reproduce a prescribed reference motion under dynamics, actuation limits, and intermittent contact. Whether an imitation is accurate is not a single geometric question. The root has to travel and turn as the reference does, the articulated body has to take the same shape, joint and root velocities have to agree, and contacts have to occur at the right moment. These objectives are measured in different units, evolve on different time scales, and interact through the dynamics, so motion imitation is a multi-objective problem whose objectives cannot be pursued independently.

The standard response is to define a tracking term for each error and to combine the terms by hand. A representative scalar reward is

$$r_t^{\text{hand}} = \sum_{g=1}^G w_g \exp\left[-\alpha_g \|\phi_g(\hat{s}_t) - \phi_g(s_t)\|_2^2\right], \quad (1)$$

where ϕ_g extracts the g -th motion feature—joint rotations, root position, velocities, end-effector positions—from the simulated state s_t and the synchronized reference state $\hat{s}_t$.

The weights w_g decide how objectives exchange against one another, and the scales α_g fix the numerical range of each error. When the motion changes, typical error magnitudes, learning difficulty, and dynamical roles change with it, so coefficients that serve one skill need not serve the next, and reward construction is repeated for every new motion.

Adversarial imitation offers a different route. Instead of declaring how many meters of root error are worth how many radians of joint error, a discriminator learns a scalar imitation signal from data. Motion priors such as AMP do this at the level of short state transitions and reward motion that resembles the dataset [7]. Adversarial differential discriminators make the tracking case precise: the discriminator receives the reference-policy differential and classifies whether it is the ideal zero vector, so the aggregation of tracking objectives is learned rather than specified [8]. The weights and scales of (1) disappear.

One coefficient remains, and it inherits the tuning burden. The differential formulation has exactly one positive example, the origin, and the optimal discriminator for a single positive point is a delta function: it scores the origin as ideal and every other differential as equally hopeless, and the policy receives no gradient. The discriminator must therefore be kept from becoming arbitrarily sharp, and the standard tool is a gradient penalty of strength λ . The coefficient is delicate in both directions, because the policy’s reward is a saturating function of the discriminator’s score. If λ is too small, the discriminator sharpens until the residuals the policy actually produces are all scored far below the ideal; the reward saturates at zero across the states the policy visits, and neither staying alive nor tracking slightly better is rewarded. If λ is too large, the discriminator is smoothed until it can no longer separate residuals that differ in the details a skill depends on; the reward is nearly the same on every state, survival is rewarded, and tracking is not. Where the usable range lies depends on the residual distribution of the motion being learned, and the published configurations accordingly use different penalty strengths for different clips and characters. The coefficient must be found again for every skill because it controls two things at once: how steep the discriminator may be anywhere, which is a property of its geometry, and how far the typical residual is pushed from the ideal, which is a property of its scale.

DARE fixes each of these with no motion-specific coefficient. The geometry is fixed by construction, using two structural facts of the differential: perfect imitation is always the same point, $\Delta = 0$, whatever the motion or phase—we call this point the anchor—and the differential is a direct sum of

semantic blocks—root position, root rotation, body position, body rotation, root velocities, joint velocities—which are the objectives of the tracking problem. Differential discriminators were conceived for general multi-objective optimization, where each objective is one scalar loss and the differential has one entry per objective, so all objectives enter on equal footing. The tracking instance replaces each entry by the raw coordinates of its objective, and the footing is lost: an 84-dimensional body-rotation block competes inside one dense layer with a 3-dimensional root-position block and wins by coordinate count. DARE restores the footing by mapping each block through its own spectrally normalized map to an equal-width latent—a learned, vector-valued generalization of the per-objective scalar loss—before a shared Lipschitz critic mixes the latents; the resulting critic changes its score, relative to the anchor, by no more than the size of the residual, for every motion and with no coefficient.

The scale is fixed by one calibration rather than by construction, because a Lipschitz bound says how steep a critic may be and not how steep it is. The balanced adversarial loss places the anchor score and the typical residual score symmetrically about zero, at $\pm M/2$ for a gap M that the loss drives as wide as the function class and the data allow. The saturating reward transform, however, responds to the score only inside a window of fixed width, and the two failure modes above are M overshooting or undershooting that window. DARE measures M once, when the input units become fixed, and rescales the critic so that anchor and typical residual sit at the two ends of the window. The rule is the same for every motion; only the measured M , like the input normalization it follows, depends on the data.

The contributions are as follows. First, we identify that the regularization coefficient of differential adversarial rewards conflates critic geometry with critic scale, and we show that this is why it must be re-tuned per motion. Second, we introduce DARE, whose factorized, spectrally normalized critic fixes the geometry by construction and whose one-shot anchor calibration fixes the scale in closed form from the reward transform, so that no reward or regularization coefficient is specified per motion. Third, we evaluate DARE with one configuration across six heterogeneous skills and execute the resulting controllers on **[TODO: robot platform]**.

II. RELATED WORK

Physics-based humanoid imitation. DeepMimic reproduces diverse motion-capture clips by combining pose, velocity, root, and end-effector tracking kernels with user-chosen coefficients [1]; later systems scale tracking through architecture and data [2]–[5] while their imitation objectives still rest on manual scales and weights. Our concern is the objective itself: a reward that can be reused across motions.

Adversarial imitation and learned rewards. GAIL casts imitation as adversarial occupancy matching [6], and AMP adapts this idea to physics-based characters by discriminating short state transitions [7]; a reward that scores “looking like the dataset” suits stylized behavior but can accept a plausible yet incomplete motion. Differential rewards hand

the reference–policy error itself to a discriminator [8]: ADD composes the tracking objectives into one differential, treats the zero vector as the sole positive example, and learns a nonlinear aggregation without component weights, stabilized by a gradient penalty whose strength is set per motion. DARE continues this program and removes that coefficient.

Stable adversarial learning. Gradient penalties, Lipschitz constraints, and spectral parameterization keep adversarial critics from becoming either brittle or saturated [9]–[11]; for losses other than the Wasserstein loss, what such a constraint controls is the range of scores the loss operates on, and the scale of that range determines whether training succeeds [13]. DARE uses the constraint for geometry alone and fixes the score range separately.

Multi-objective optimization. Weighted sums, Pareto methods, adaptive loss balancing, and multi-critic RL all address competing objectives and typically require explicit preferences or multiple objective gradients [14]–[16]; differential adversarial learning instead learns a nonlinear scalarization from data, and DARE performs it with one critic and one configuration for every motion.

III. BACKGROUND

A. Reference-Conditioned Humanoid Control

Let s_t be the simulated state at time t , $\hat{s}_t$ the synchronized reference state, and a_t the action, a vector of desired joint positions that a proportional–derivative servo converts into torques at the control frequency. A policy π_φ observes proprioception together with the next few reference frames and must close the loop under contact and underactuation; episodes begin at a random reference phase, so the policy has to recover the clip from any point. The policy maximizes the discounted return of a reward r_t with PPO [17], [18]; Section IV constructs r_t .

B. Motion Tracking as Multi-Objective Optimization

A motion descriptor decomposes a state into G feature groups $\phi_g : \mathcal{S} \rightarrow \mathbb{R}^{d_g}$, each describing one physical aspect of the motion: global root position, root rotation, root-relative body positions, body rotations, root linear and angular velocity, and joint velocity. A handcrafted reward first compresses each group into a distance $e_{g,t} = \|\phi_g(\hat{s}_t) - \phi_g(s_t)\|_2$ and then applies the fixed map (1). The widely used DeepMimic reward does this with five terms whose weights (0.5, 0.1, 0.15, 0.1, 0.15) and scales (0.25, 0.01, 5, 1, 10) were selected by hand, together with fourteen joint-specific weights inside the pose and velocity terms [1], [8]: twenty-four coefficients, each encoding an assumption about how large an error of one kind is worth relative to another, frozen before the residual distribution of any policy is seen. A learned scalarization can instead adapt the trade-off to the residuals the current policy actually produces.

C. Learned Differential Objectives

Differential adversarial learning provides that scalarization [8]. In its general form, n objectives with non-negative

losses $l^1, \dots, l^n$ are collected into a differential vector $\Delta = (l^1, \dots, l^n)$ that records the gap between ideal and actual performance on every objective. The ideal solution is the single point $\Delta = 0$. A discriminator $D(\Delta)$ trained to separate this point from the differentials of the current model is a learned, nonlinear aggregation of the objectives, and its gradient $\partial D / \partial l^i$ is the weight the aggregation currently assigns to objective i ; that weight moves during training toward the objectives on which the model still falls short. For synchronized motion imitation the same construction is applied to the feature differential $\phi(\hat{s}_t) - \phi(s_t)$. Perfect tracking maps every pair onto one origin, so the positive class is again a single known point, and the differential retains information about *how* the current motion departs from the reference rather than collapsing it into a state occupancy. The policy is rewarded at each step with $-\log(1 - D(\Delta_t))$, as in AMP: large when the discriminator finds the residual Δ_t close to ideal, and vanishing when it does not.

A single positive point has a consequence. Without regularization the optimal discriminator is a delta function at the origin, which scores every policy residual as equally hopeless, so the reward vanishes on every state the policy visits and provides no gradient. The discriminator is therefore trained with a gradient penalty $\lambda \mathbb{E}_{\Delta \sim q_\pi} \|\nabla_\Delta D(\Delta)\|_2^2$ added to the classification objective, where q_π denotes the distribution of policy residuals. The penalty acts on where those residuals lie, and a fixed λ yields a differently shaped discriminator on every motion. Small λ lets the discriminator approach the delta function and saturate the reward; large λ smooths it until residuals that differ in skill-relevant detail receive the same score. The published configurations bear this out: λ is set to 0.1 for the Climb clip and to 1 for the other humanoid clips, and to different values again for a biped, a quadruped, and a planar walker, while removing the penalty altogether fails [8]. How those values were chosen is not reported; what they document is that no single value served all settings. Locating the window between the two failures is the tuning that the learned aggregation was meant to remove; the next section shows that λ is asked to do two separable jobs and does each without it.

IV. DARE: DIFFERENTIAL ADVERSARIAL REWARD EQUALIZATION

A. Learned Differential Reward

The distances $e_{g,t}$ are convenient for a weighted sum, but they discard the signed structure of the residual. Two policies can share the same root-position distance while drifting in opposite directions, and only one of them remains compatible with the next contact. DARE therefore keeps the differential itself. For each group g , a normalizer $\mathcal{N}_g = \text{diag}(\nu_g)^{-1}$ rescales every coordinate by its mean absolute value ν_g , estimated from the residuals of the first 10^8 training samples and then fixed, as is standard for adversarial motion rewards. The normalizer has no offset, so it removes unit disparity between meters, rotation coordinates, and velocities while leaving the origin fixed; rotations use a tangent-normal

embedding so that differences are Euclidean. The group-wise residual, the full differential, and the ideal point are

$$\Delta_{g,t} = \mathcal{N}_g(\phi_g(\hat{s}_t) - \phi_g(s_t)), \quad \Delta_t = \bigoplus_{g=1}^G \Delta_{g,t}, \quad \Delta^+ = 0. \quad (2)$$

The vector $\Delta_t \in \mathbb{R}^d$, $d = \sum_g d_g$, retains error direction and cross-coordinate relationships instead of G distances, and every perfectly synchronized pair maps to the same origin regardless of the motion or its phase. Differentials produced by the current policy, together with recently visited differentials kept in a replay buffer, form the distribution q_π of non-ideal residuals.

The remaining question is how to turn Δ into a scalar the policy can maximize. Rather than assigning a price to each block, we ask a critic to recognize the ideal point. Let $D_\theta(\Delta) = \sigma(f_\theta(\Delta))$ be the probability that a differential is ideal, where σ is the logistic sigmoid and $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}$ is a scalar score. The origin is the positive class and residuals drawn from q_π are the negative class, so the critic plays the usual two-player game,

$$\max_{\theta} \log D_\theta(0) + \mathbb{E}_{\Delta \sim q_\pi} [\log(1 - D_\theta(\Delta))]. \quad (3)$$

The first term asks the critic to accept perfect imitation; the second asks it to reject the residuals of the current controller and, through the replay in q_π , its recent history. Maximizing (3) is the same as minimizing the balanced Bernoulli negative log-likelihood of the labeling; writing $-\log \sigma(z) = \text{softplus}(-z)$ and $-\log(1 - \sigma(z)) = \text{softplus}(z)$ puts it in the form we work with,

$$\mathcal{L}_{\text{DARE}}(\theta) = \frac{1}{2} \text{softplus}[-f_\theta(0)] + \frac{1}{2} \mathbb{E}_{\Delta \sim q_\pi} \text{softplus}[f_\theta(\Delta)]. \quad (4)$$

This is the differential objective of Section III-C without its penalty. The labels state only whether a differential is ideal; the relative importance of root, pose, and velocity residuals is left to f_θ , which estimates it from the residual distribution of the current policy and re-estimates it as q_π concentrates; this adaptive aggregation is what replaces the manual scalarization (1).

The critic's verdict is the reward. $D_\theta(\Delta_t)$ is the probability that the residual at step t is ideal, so the policy should be paid the more, the smaller $1 - D_\theta(\Delta_t)$ is; the conversion used by AMP and ADD implements exactly this, and in terms of the score it is a softplus:

$$r_t^{\text{DARE}} = -\beta \log[1 - D_\theta(\Delta_t)] = \beta \text{softplus}[f_\theta(\Delta_t)], \quad (5)$$

with a fixed amplitude β shared by every skill. The reward is non-negative, so it also pays the policy for staying alive, and it grows without bound as the critic becomes more confident that the residual is ideal; the same f_θ aggregates every tracking objective, so the policy never consults a component distance, a success threshold, or a component coefficient. What keeps (4) from producing the delta function—and (5) from collapsing to zero on every state the policy visits—is not a penalty but the function class of f_θ ; the next two subsections choose that class so that its geometry treats the objectives fairly and its scale suits the reward.

B. Equalized Critic: Geometry by Construction

In the general differential of Section III-C, every objective contributes one entry. For the g -th motion feature that entry is the distance $l^g = \|\Delta_g\|_2$: a scalar that is 1-Lipschitz in Δ_g and vanishes exactly at the ideal point, and whose partial derivative is the objective's weight. The tracking differential drops the scalar and hands the critic the d_g coordinates of Δ_g directly, and a dense first layer treats them as follows. After per-coordinate normalization every coordinate has unit typical magnitude, so $\|\Delta_g\|_2$ grows like $\sqrt{d_g}$: an 84-dimensional body-rotation residual is about five times longer than a 3-dimensional root-position residual of the same per-coordinate size. Suppose the first layer is a single dense map $W = [W_1 \cdots W_G]$ whose gain is bounded, $\sigma_{\max}(W) \leq 1$ —the constraint we impose on every layer below to obtain the geometric guarantee. Along any unit output direction u ,

$$\sum_{g=1}^G \|W_g^T u\|_2^2 = \|W^T u\|_2^2 \leq 1, \quad (6)$$

so the G blocks share one unit of gain in every output unit, and gradient descent allocates that unit in proportion to the raw norms of the blocks. The critic sees anonymous coordinates rather than objectives, and the weights it learns follow the feature layout rather than the skill. Root tracking is indeed where differential discriminators have been reported to be least precise [8].

DARE restores the per-objective structure without returning to a scalar. Each factor $\Delta_g \in \mathbb{R}^{d_g}$ is mapped by its own affine map and a 1-Lipschitz activation ρ to an equal-width latent, and the latents are concatenated:

$$\begin{aligned} \Phi_{\oplus}(\Delta) &= \bigoplus_{g=1}^G \psi_g(\Delta_g), \\ \psi_g(\Delta_g) &= \rho(\bar{W}_g \Delta_g + b_g), \quad \bar{W}_g = \frac{W_g}{\sigma_{\max}(W_g)}, \end{aligned} \quad (7)$$

where $W_g \in \mathbb{R}^{m \times d_g}$ and every factor receives the same latent width m . Each ψ_g is 1-Lipschitz in Δ_g , exactly as l^g was; with $m = 1$ and W_g fixed to a unit row it reduces to a scalar loss, and with $m > 1$ and learned W_g it is a vector-valued loss that keeps the direction of the residual and adapts to the geometry of its block. The direct sum keeps each ψ_g in its own slot, so what follows receives one equal-width entry per objective, as in the general form. A shared trunk then scores the concatenation, $g_{\theta}(\Delta) = \bar{T}(\Phi_{\oplus}(\Delta))$ with $\bar{T}(h) = \bar{w}^T \rho(\bar{V}h + c) + b$, whose linear maps are spectrally normalized in the same way so that $\text{Lip}(\bar{T}) \leq 1$. We write g_{θ} for this raw critic output; the score f_{θ} that enters the loss (4) and the reward (5) is g_{θ} up to a single scalar fixed in Section IV-C, and equals g_{θ} until then. All affine maps carry a bias, so activation thresholds can sit where the residual distribution actually lives; the anchor $g_{\theta}(0)$ thus depends on every layer and is trained by the positive sample like any other point, while spectral normalization keeps each map non-expansive regardless of the offsets.

Equalization gives every objective the same latent capacity and the same gain bound before interaction. In contrast to (6), each factor in (7) owns its output coordinates and attains $\sigma_{\max}(\bar{W}_g) = 1$ independently of the others. Because the blocks act on disjoint outputs, their gains add in quadrature instead of competing: for any two differentials,

$$\begin{aligned} \|\Phi_{\oplus}(\Delta) - \Phi_{\oplus}(\Delta')\|_2^2 &= \sum_{g=1}^G \|\psi_g(\Delta_g) - \psi_g(\Delta'_g)\|_2^2 \\ &\leq \|\Delta - \Delta'\|_2^2, \end{aligned} \quad (8)$$

since each block has Lipschitz constant at most one and the norm of a concatenation is the root-sum of the block norms. The Jacobian of $\Phi_{\oplus}$ is block diagonal across factors, so a perturbation in one semantic subspace cannot be amplified through the coordinates of another, and all G factors attain unit gain simultaneously while the layer as a whole stays non-expansive. The weight of objective g , $\omega_g = \|\partial g_{\theta} / \partial \Delta_g\|_2$, is then bounded by the trunk's sensitivity to its slot, $\omega_g \leq \|\partial g_{\theta} / \partial h_g\|_2$ with $h_g = \psi_g(\Delta_g)$. The trunk still divides one unit of sensitivity among the slots, but that division is now learned from the residual distribution rather than dictated by coordinate count: the first layer no longer lets a long block crowd out a short one before the trunk has seen either. Equalization equalizes capacity and gain; it does not equalize the norms $\|\Delta_g\|$, which are the sufficient statistics of each block's evidence and are left intact.

The unique ideal point now gives the geometric guarantee. Because every perfect pair occupies the origin, the quantity that governs reward variation is the score change relative to the anchor, and combining (8) with $\text{Lip}(\bar{T}) \leq 1$ gives

$$|g_{\theta}(0) - g_{\theta}(\Delta)| \leq \|\Delta\|_2. \quad (9)$$

If the realized motion changes only slightly relative to the reference, the score changes by no more than the normalized residual; the biases cancel in every difference, so they move the critic's activation boundaries without loosening the bound. This is the geometric job of the gradient penalty in Section III-C—nearby residuals must not receive arbitrarily different scores—done by the parameterization: it holds at initialization, at every training step, and for every motion, with no coefficient.

C. Anchor Calibration: Scale in Closed Form

The bound (9) says how steep the critic may be. It does not say how steep the critic is, and the policy is sensitive to the difference, because it receives the score through the fixed transform (5). How much that reward responds to a change in the score is its slope,

$$\frac{\partial r^{\text{DARE}}}{\partial f_{\theta}} = \beta \sigma(f_{\theta}), \quad (10)$$

a sigmoid that rises from 0 for very negative scores to β for very positive ones. Only over the transition of this sigmoid does a change in the score change the reward. Far below the transition the reward is zero however the residual changes; far above it the reward is linear in the score and the transform has

stopped shaping anything. The critic is useful to the policy only where its scores fall on the transition.

Where they fall is decided by the loss, not by the bound. The output bias of the critic shifts every score by the same amount, and the loss (4) is stationary with respect to that shift when its two terms have equal slope, $\mathbb{E}_{q_\pi} \sigma(f_\theta(\Delta)) = \sigma(-f_\theta(0))$. The left-hand side is the mean probability of ideality the critic assigns to policy residuals, and the score whose sigmoid equals that mean is the natural representative of the negative class, $\bar{f}_\theta = \sigma^{-1}(\mathbb{E}_{q_\pi} \sigma(f_\theta(\Delta)))$, which we call the typical residual score. In these terms the stationarity condition reads $\bar{f}_\theta = -f_\theta(0)$: writing $M = f_\theta(0) - \bar{f}_\theta$ for the anchor gap,

$$f_\theta(0) = +\frac{M}{2}, \quad \bar{f}_\theta = -\frac{M}{2}. \quad (11)$$

The anchor and the typical residual sit symmetrically about zero, and the loss pushes M as wide as the function class and the data permit. ($\bar{f}_\theta$ coincides with the arithmetic mean of the scores to second order in their spread; the runs reported below computed M with the arithmetic mean, which differed from $\bar{f}_\theta$ by **TODO**% at t_f .) The bound (9) caps M by the size of the residuals themselves; the realized value is some fraction of that cap, set by how much of the bound the trained network uses and by the residual scale of the motion, neither of which is a design quantity. The two failure modes of Section III-C are the two placements of M relative to the transition of (10). If M is too large, the typical residual at $-M/2$ lies far below the transition: its reward and its reward slope are both near zero, staying alive earns nothing, and improving a poor residual earns nothing either. If M is too small, the ratio of the reward slope at the anchor to that at the typical residual, $\sigma(M/2)/\sigma(-M/2) = e^{M/2}$, approaches one: the reward is close to the constant $\beta \log 2$ on every state, survival is rewarded, and the differences that distinguish good tracking from bad are compressed. A gradient penalty moves M only indirectly, through the critic's slope on sampled residuals, so the same λ lands M in different places for different motions. DARE sets M directly.

Both endpoints must therefore lie on the part of the transition where the slope is neither negligible nor saturated, and the question is how wide that part is. A sigmoid has no sharp edges; the conventional way to give its transition a width—the same convention used for the rise time of any sigmoidal response—is the span over which it climbs from 20% to 80% of its maximum. For the slope (10) this span is

$$w = \sigma^{-1}(0.8) - \sigma^{-1}(0.2) = \ln 4 - (-\ln 4) = \ln 16 \approx 2.77. \quad (12)$$

Placing the anchor at the upper end of the transition and the typical residual at the lower end means, by (11), $M = w$. The choice has a direct reading in terms of the critic itself: $D_\theta(0) = \sigma(\ln 4) = 0.8$ and $D_\theta(\bar{\Delta}) = 0.2$, so the calibrated critic assigns the anchor an 80% probability of being ideal and the typical residual 20%—a confident but not categorical classifier. A larger M drives it toward a hard classifier whose reward vanishes on typical residuals, a smaller M toward a coin flip whose reward carries no information; keeping

an adversarial critic in a band of moderate confidence is standard practice [12]. The convention is fixed a priori and never selected by task outcome, and it passes an independent check: on the hardest skill in our benchmark (Section V-C), an uncalibrated critic that succeeded reached a gap at t_f within 2% of $\ln 16$, without either quantity having been fitted to the other.

DARE achieves $M = w$ with one scalar applied to the raw critic g_θ of Section IV-B:

$$f_\theta(\Delta) = \kappa g_\theta(\Delta), \quad \kappa = \frac{w}{M_f}, \quad (13)$$

$$M_f = g_\theta(0) - \sigma^{-1}(\mathbb{E}_{q_\pi} \sigma(g_\theta(\Delta))) \Big|_{t=t_f}.$$

The gap M_f is measured once, at the iteration t_f at which the input normalizer $\mathcal{N}$ is fixed, over the same mixture of current and replayed residuals that forms the negative class of (4); κ is held constant thereafter, and before t_f it equals one. The moment is forced by the units: before t_f the coordinates of Δ are still being rescaled and M has no stable meaning. Both the loss (4) and the reward (5) use the calibrated f_θ , so the critic is trained and read out at one scale.

Three properties follow. The anchor is untouched: $\kappa \cdot 0 = 0$, and $f_\theta(0) = \kappa g_\theta(0)$ is still trained by the positive sample through every layer. The geometry is untouched: κ rescales the critic uniformly, so the ordering of residuals is preserved and (9) becomes $|f_\theta(0) - f_\theta(\Delta)| \leq \kappa \|\Delta\|_2$, a Lipschitz constant derived from the data rather than assumed to be one. And the rule contains no motion-specific number: w is a constant of the reward transform and M_f a statistic of the training data, so, like the normalizer $\mathcal{N}$, κ takes a different value on every motion because the same formula meets different residuals. Architecture, optimizer, replay, and reward amplitude remain fixed choices inherited from AMP and ADD; what DARE removes is the coefficient that had to be chosen per motion.

D. Training

Training alternates rollout collection, differential construction, critic updates, and PPO updates (Algorithm 1). Critic minibatches draw an equal mixture of current and replay differentials, with the ideal point presented once per minibatch as the positive class; at t_f , before the rewards of that iteration are computed, M_f is measured with the critic in evaluation mode and κ is set by (13). One critic, one reward, one policy, and one calibration rule make up the whole loop, and every quantity in it is shared across motions.

V. EXPERIMENTS

A. Setup

All methods control the 28-DoF, 15-body simulated humanoid of DeepMimic and ADD with PD joint targets at 30 Hz in Isaac Lab (120 Hz physics) [19], [20]. Episodes start at a random reference phase and terminate on undesired ground contact; DeepMimic, ADD, and DARE use pose termination with a unit threshold, AMP follows its standard protocol without it. Each policy is trained for 2000 iterations

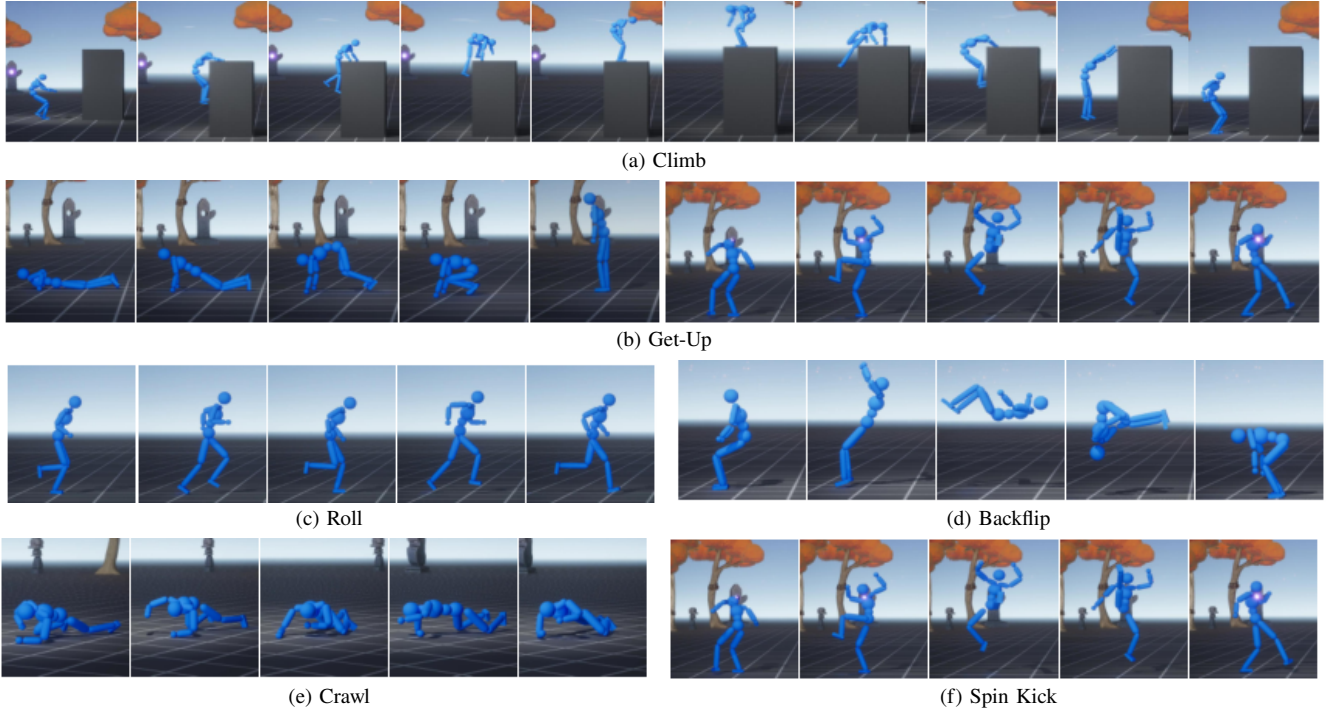


Fig. 1. DARE policies executing the six skills on hardware [TODO: platform]. Frames are equally spaced in time; the two longer clips are shown with ten frames, the four shorter clips with five. One configuration produced every controller shown. [TODO: All strips are placeholders cropped from the ADD paper’s Fig. 2 and must be replaced by our own robot photographs, with the reference motion overlaid or shown alongside.]

Algorithm 1 DARE Training

Require: reference motion $\mathcal{M}$, policy π_φ , value V_ω , raw critic g_θ , replay buffer $\mathcal{B}$, normalizer $\mathcal{N}$; $\kappa \leftarrow 1$

- 1: **for** iteration $t = 1, 2, \dots$ **do**
- 2: Collect a synchronized rollout with π_φ from random reference phases; construct Δ_t by (2); add to $\mathcal{B}$
- 3: **if** $\mathcal{N}$ was fixed this iteration **then**
- 4: $\bar{g} \leftarrow \sigma^{-1}(\mathbb{E}_{\mathcal{B}_{\text{rollout}}} \sigma(g_\theta(\Delta)))$
- 5: $M_f \leftarrow g_\theta(0) - \bar{g}$; $\kappa \leftarrow \ln 16/M_f$
- 6: **end if**
- 7: Compute r_t^{DARE} by (5) with $f_\theta = \kappa g_\theta$
- 8: **for** critic minibatch **do**
- 9: Sample equal numbers of current and replay differentials; update g_θ by one step on (4) with $f_\theta = \kappa g_\theta$
- 10: **end for**
- 11: Update π_φ and V_ω with PPO on r_t^{DARE}
- 12: **end for**

of 8192 environments $\times$ 32 steps (5.24×10^8 samples); the input normalizer is fixed after 10^8 samples, at iteration 382, where DARE calibrates. Actor and value networks are two-layer MLPs (1024, 512); PPO settings (clip 0.2, GAE 0.95, $\gamma = 0.99$, SGD with momentum 0.9) are shared by all learned-reward methods.

The six skills span distinct dynamical regimes: cyclic ground rotation (Roll, 2.00 s), long-horizon object interaction (Climb, 9.97 s), aerial inversion (Backflip, 1.75 s), prone locomotion (Crawl, 2.93 s), recumbent recovery (Get-Up, 3.03 s), and a ballistic kick (Spin Kick, 1.28 s). We report position tracking error in meters—the root displacement and the root-relative joint displacements averaged over joints and frames—and DoF-velocity error in radians per second, the

TABLE I

DARE CONFIGURATION, IDENTICAL FOR ALL SKILLS. THE CALIBRATED SCALE κ IS NOT A SETTING BUT THE OUTPUT OF (13).

Component	Setting
Differential	single frame, $d = 172$
Semantic factors	$G=7$: root/body pos. (3/45), root/body rot. (6/84), root lin./ang. vel. (3/3), DoF vel. (28)
Direct-sum layer	$m = 146$ per factor (1022 total)
Shared trunk	linear $1022 \rightarrow 512$; output $512 \rightarrow 1$
Activation ρ	ReLU (1-Lipschitz)
Spectral norm.	all linear maps, one power iteration
Normalizer $\mathcal{N}$	per-coordinate mean-abs scale, fixed after 10^8 samples (iter. 382)
Calibration	$\kappa = \ln 16/M_f$ at that moment, once
Critic optimizer	SGD, lr 2.5×10^{-4} , momentum 0.9
Critic minibatch	16384 current + 16384 replay
Replay buffer	2×10^5 differentials
Reward amplitude β	2

two quantities used by prior humanoid tracking studies [8]. In Table II the AMP, DeepMimic, and ADD entries are the published results on the same clips and metrics; the curves in Fig. 2 come from our own runs of all four methods under the protocol above, each baseline with its published training settings, and every DARE run uses Table I. Each method is trained with five seeds and evaluated on 4096 episodes from random reference phases per seed (mean $\pm$ std over seeds). Because a frame error can stay moderate while the motion is never performed, we also report completion: an

TABLE II

TRACKING PERFORMANCE ON SIX HETEROGENEOUS SKILLS. DARE USES ONE CONFIGURATION ON EVERY ROW. AMP, DEEPMIMIC, AND ADD ARE THE RESULTS PUBLISHED BY ZHANG *ET AL.* ON THE SAME CLIPS AND METRICS [8]; COMPLETION IS THE FRACTION OF DARE EPISODES THAT SATISFY THE SKILL-SPECIFIC CRITERION OF SECTION V-A. MEAN $\pm$ STD OVER FIVE SEEDS; LOWER ERROR IS BETTER. [TODO: DARE COLUMNS: FIVE-SEED, RANDOM-START EVALUATION; THE CLIMB AND ROLL ENTRIES SHOWN ARE SINGLE-SEED, PHASE-0 PRELIMINARY VALUES.]

Skill	Len. (s)	Position [m] ↓				DoF vel. [rad/s] ↓				Compl. [%]
		AMP	DeepMimic	ADD	DARE	AMP	DeepMimic	ADD	DARE	DARE
Roll	2.00	.141 $\pm$.031	.115 $\pm$.132	.152 $\pm$.005	.047	1.576 $\pm$.318	.994 $\pm$.051	1.330 $\pm$.101	.897	100
Climb	9.97	.185 $\pm$.021	.066 $\pm$.004	.046 $\pm$.031	.029	1.190 $\pm$.012	.567 $\pm$.015	.374 $\pm$.071	.496	100
Backflip	1.75	.267 $\pm$.015	.111 $\pm$.054	.062 $\pm$.001	TODO	2.243 $\pm$.113	1.103 $\pm$.024	.878 $\pm$.013	TODO	TODO
Crawl	2.93	.050 $\pm$.006	.027 $\pm$.000	.028 $\pm$.002	TODO	.646 $\pm$.089	.430 $\pm$.006	.283 $\pm$.002	TODO	TODO
Get-Up	3.03	.096 $\pm$.018	.023 $\pm$.001	.022 $\pm$.001	TODO	.838 $\pm$.029	.433 $\pm$.008	.325 $\pm$.005	TODO	TODO
Spin Kick	1.28	.064 $\pm$.010	.078 $\pm$.062	.025 $\pm$.000	TODO	1.453 $\pm$.327	1.222 $\pm$.233	.774 $\pm$.007	TODO	TODO

episode counts as complete if it survives to the end of the clip and passes a skill-specific test fixed in advance—one full signed revolution of the root for Roll; ascent, descent, and foot clearance of the box for Climb; height gain and an upright landing for Backflip; forward progress for Crawl; a standing final height for Get-Up; root yaw progress for Spin Kick. [TODO: confirm the Spin Kick criterion against the evaluation code]

[TODO: Hardware paragraph, to be written once deployment is done: name the robot and its degrees of freedom; how the six reference clips were retargeted to it; that the same DARE configuration of Table I was used to train the deployed policies on the robot’s model without any change to the reward or the calibration rule; the sim-to-real measures (domain randomization, observation noise, action filtering) and the control frequency on hardware; and how many trials per skill were executed and how many completed.]

B. Cross-Motion Tracking

Table II asks whether one configuration can serve skills whose contacts, horizons, and velocity profiles differ, and Fig. 1 shows the resulting controllers executing the six skills on hardware. Roll is a cyclic ground skill in which root travel and body rotation must stay in phase; a policy that lies down and stands up again achieves a moderate frame error while missing the motion entirely, and this is the reported failure mode of differential discriminators on this clip [8]. Climb is a ten-second sequence of hand and foot contacts with a box, where an early slip forecloses the rest of the clip; Backflip demands a precise landing velocity, Crawl turns small orientation errors into large position errors, Get-Up must discover a contact sequence that restores stance, Spin Kick is short and ballistic.

On Roll, DARE completes the revolution (winding ratio 1.02, phase alignment 1.0) with a position error of 0.047 m, less than half the best published value for this clip. On Climb, all evaluated episodes complete the ascent and descent, with position error 0.029 m against the published 0.046 m and DoF-velocity error 0.496 rad/s against 0.374 rad/s. [TODO: Replace with five-seed, random-start statistics; report the remaining four skills; state on how many skills DARE matches or improves the best published tracking error and where

the equalized differential helps most; compare seed spreads with DeepMimic on Roll, Backflip, and Spin Kick, where handcrafted rewards show wide seed bands in the published results.] [TODO: Hardware results: trials and completions per skill on the robot, and which skills transferred without adjustment.]

C. Operating Point and Learning Curves

Figure 2(a) shows the anchor gap M over training: before t_f each skill’s critic separates the anchor from its residuals by a different amount, at t_f the rule brings every skill to $\ln 16$, and afterwards the gaps contract together as the residuals shrink toward the anchor. [TODO: Report the range of uncalibrated gaps at t_f across the six skills, and the fraction $M_f/\mathbb{E}\|\Delta\|$ of the bound the trained critic uses (about 0.2–0.3 on Climb in preliminary measurements).] On the two skills completed so far the uncalibrated gap was within 2% of $\ln 16$ on Climb and 13% above it on Roll, and by the end of training both had settled to the same typical-residual score, -1.01 : the scale fixed at t_f is thereafter motion-agnostic in the reward slope on the states the policy visits. Panel (b) plots position error against environment samples under a matched budget; solid curves are seed means, shaded regions standard deviations. [TODO: Describe convergence speed, final plateau, and seed consistency of DARE relative to AMP, DeepMimic, and ADD.]

D. Ablation

DARE makes two structural choices about the critic’s geometry, the direct-sum first layer and the spectral parameterization of the trunk, and Table III varies each on Roll and Climb, representative short-cycle and long-horizon regimes. All variants share the actor, PPO, budget, differential (2), and critic width, and all are calibrated by (13), so the rows differ in geometry alone; the scale that calibration sets, and what it would have been without it, is read from the pre-calibration gaps in Fig. 2(a). *Flat+SN* replaces the direct-sum layer by one spectrally normalized dense map of the same output width, so the first nonlinearity already mixes every coordinate; it tests whether factorization before interaction matters. *Factorized+GP* keeps the direct-sum layer but replaces the spectral parameterization of the trunk by a gradient penalty with one coefficient shared by both skills; it tests whether the

[TODO: Figure, two panels. (a) Anchor gap M versus iteration for the six skills (one curve each); mark t_f and the level $\ln 16$. Before t_f the curves sit at skill-dependent heights, at t_f they all land on the line, afterwards they contract together. (b) Position tracking error versus environment samples for AMP, DeepMimic, ADD, and DARE on Roll and Climb; seed means with standard-deviation bands; matched budget.]

Fig. 2. (a) Where the critic operates: calibration places the anchor and the typical residual at the ends of the reward slope’s transition at t_f , whatever the skill. (b) Learning curves on two representative regimes.

TABLE III

ABLATION ON TWO REPRESENTATIVE SKILLS. MEAN $\pm$ STD OVER FIVE SEEDS;
LOWER IS BETTER. [TODO: FILL FROM THE FORMAL ABLATION RUNS]

Variant	Position [m] $\downarrow$		DoF vel. [rad/s] $\downarrow$	
	Roll	Climb	Roll	Climb
Flat+SN	TODO	TODO	TODO	TODO
Factorized+GP	TODO	TODO	TODO	TODO
DARE	TODO	TODO	TODO	TODO

architectural bound does more than a coefficient-controlled penalty of the same intent.

The Flat+SN row is the direct test of Section IV-B: (6) predicts that the single spectral budget of the dense map is consumed by the long blocks, so the critic prices coordinate cardinality rather than contact, balance, or timing. [TODO: Report the measured degradation and whether it is larger on Roll, where root travel and body rotation compete directly.] The Factorized+GP row keeps the semantic channels and controls sensitivity with a coefficient instead of through the maps. [TODO: Report.]

VI. CONCLUSION

Manual tracking rewards turn the aggregation of heterogeneous imitation objectives into per-motion engineering, and learned differential rewards, having removed the weights, leave one coefficient that must still be found per motion because it decides two things at once: how steep the critic may be and how far the typical residual is pushed from the anchor. DARE decides each separately, with no coefficient chosen per motion. Every objective enters through its own spectrally normalized slot, so a three-dimensional root residual and an eighty-four-dimensional rotation residual stand on equal footing and the score change relative to the unique zero differential is bounded by the residual; the critic is then rescaled once, in closed form, so that the anchor and the typical residual sit at the two ends of the reward slope’s transition. [TODO: One-sentence quantitative summary of the six-skill results and the ablation.] Extending DARE to multi-clip datasets and to other morphologies requires no change to the rule, and the factorized critic offers a direct place to attach task objectives as further semantic blocks.

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